```
Microsoft Windows [Version 10.0.26100.7309]
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C:\Users\Lenovo>cd\

C:\>cd xampp

C:\xampp>cd mysql

C:\xampp\mysql>cd bin

C:\xampp\mysql\bin>mysql -u root -p -h 127.0.0.1
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.32-MariaDB mariadb.org binary distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

1) create a database named customers_db

```
MariaDB [(none)]> create database customer_db;
Query OK, 1 row affected (0.006 sec)

MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| customer_db        |
| eemc_db            |
| information_schema |
| mysql              |
| performance_schema |
| phpmyadmin         |
| test               |
+--------------------+
7 rows in set (0.048 sec)
```

```
MariaDB [(none)]> use customer_db;
Database changed
```

2) create a table named customers with following columns adding constraints following constraints

Customer_id: Primary key

Name: not null

customer_id	Name	Email	Age	Address
Int	varchar(30)	varchar(30)	Int	varchar(30)

```
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | int(11)      | NO   | PRI | NULL    |       |
| name        | varchar(30)  | NO   |     | NULL    |       |
| email       | varchar(255) | YES  |     | NULL    |       |
| age         | int(11)      | YES  |     | NULL    |       |
| address     | varchar(30)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
```

3) We have already added two constraints (PRIMARY KEY and NULL) while creating table customers. We can add constraints after creating table as well.
Now add the following constraint to the above table named customers

- ✓ Email must be unique of each customer

```
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | int(11)      | NO   | PRI | NULL    |       |
| name        | varchar(30)  | NO   |     | NULL    |       |
| email       | varchar(255) | YES  | UNI | NULL    |       |
| age         | int(11)      | YES  |     | NULL    |       |
| address     | varchar(30)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
```

- ✓ Age must be greater than 18

```
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| customer_id | int(11)      | NO   | PRI | NULL    |       |
| name        | varchar(30)  | NO   |     | NULL    |       |
| email       | varchar(255) | YES  | UNI | NULL    |       |
| age         | int(11)      | YES  |     | NULL    |       |
| address     | varchar(30)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
```

- ✓ The default value of address must be bhaktapur.

```
+-------------+--------------+------+-----+-----------+-------+
| Field       | Type         | Null | Key | Default   | Extra |
+-------------+--------------+------+-----+-----------+-------+
| customer_id | int(11)      | NO   | PRI | NULL      |       |
| name        | varchar(30)  | NO   |     | NULL      |       |
| email       | varchar(255) | YES  | UNI | NULL      |       |
| age         | int(11)      | YES  |     | NULL      |       |
| address     | varchar(30)  | YES  |     | bhaktapur |       |
+-------------+--------------+------+-----+-----------+-------+
```

4) Now insert records of customer with and without violating constraints.
PRIMARY KEY

```
MariaDB [customer_db]> insert into customers values(1,'ram','ram@gmail.com',22,'kathmandu');
Query OK, 1 row affected (0.046 sec)

MariaDB [customer_db]> insert into customers values(2,'sita','sita@gmail.com',25,'pokhara');
Query OK, 1 row affected (0.006 sec)

MariaDB [customer_db]> select * from customers;
+-------------+------+----------------+------+-----------+
| customer_id | name | email          | age  | address   |
+-------------+------+----------------+------+-----------+
|           1 | ram  | ram@gmail.com  |   22 | kathmandu |
|           2 | sita | sita@gmail.com |   25 | pokhara   |
+-------------+------+----------------+------+-----------+
```

Insertion can be performed successfully.

```
MariaDB [customer_db]> insert into customers values(2,'gita','gita@gmail.com',26,'butwal');
ERROR 1062 (23000): Duplicate entry '2' for key 'PRIMARY'
```

This cannot inserted because customer_id is primary key which should be unique and not null

NOT NULL

```
MariaDB [customer_db]> select * from customers;
+-------------+------+----------------+------+-----------+
| customer_id | name | email          | age  | address   |
+-------------+------+----------------+------+-----------+
|           1 | ram  | ram@gmail.com  |   22 | kathmandu |
|           2 | sita | sita@gmail.com |   25 | pokhara   |
|           3 |      | gita@gmail.com |   26 | butwal    |
+-------------+------+----------------+------+-----------+
```

CHECK

```
MariaDB [customer_db]> insert into customers values (4,'hari','hari@gmail.com',27,'lalitpur');
Query OK, 1 row affected (0.007 sec)

MariaDB [customer_db]> select * from customers;
+-------------+------+----------------+------+-----------+
| customer_id | name | email          | age  | address   |
+-------------+------+----------------+------+-----------+
|           1 | ram  | ram@gmail.com  |   22 | kathmandu |
|           2 | sita | sita@gmail.com |   25 | pokhara   |
|           3 |      | gita@gmail.com |   26 | butwal    |
|           4 | hari | hari@gmail.com |   27 | lalitpur  |
+-------------+------+----------------+------+-----------+
```

// this will works

```
MariaDB [customer_db]> insert into customers values (11,'ganesh','ganesh@gmail.com',15,'dolkha');
ERROR 4025 (23000): CONSTRAINT `age_check` failed for `customer_db`.`customers`
```

//this will not works because age must be greater than 18

UNIQUE

```
MariaDB [customer_db]> insert into customers values (16,'ramesh','ram@gmail.com',24,'chitwan');
ERROR 1062 (23000): Duplicate entry 'ram@gmail.com' for key 'email'
```

//this will not works because Duplicate entry 'ram@gmail.com' for key 'email'

DEFAULT

```
MariaDB [customer_db]> insert into customers(customer_id,name,email,age) values(5,'gopal','gopal@gmail.com',20);
Query OK, 1 row affected (0.007 sec)

MariaDB [customer_db]> select * from customers;
+-------------+-------+----------------+------+-----------+
| customer_id | name  | email          | age  | address   |
+-------------+-------+----------------+------+-----------+
|           1 | ram   | ram@gmail.com  |   22 | kathmandu |
|           2 | sita  | sita@gmail.com |   25 | pokhara   |
|           3 |       | gita@gmail.com |   26 | butwal    |
|           4 | hari  | hari@gmail.com |   27 | lalitpur  |
|           5 | gopal | gopal@gmail.com |   20 | bhaktapur |
+-------------+-------+----------------+------+-----------+
```

FOREIGN KEY

We have already created table **customers** with following columns

customer_id	Name	Email	age	address
Int	varchar(30)	varchar(30)	Int	varchar(30)

Now we have to create table **orders** with following columns.

Order_id	Product_name	Price	Customer_id

and **order_id** is primary key and **customer_id** as foreign key for customer

```
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| order_id    | int(11)     | NO   | PRI | NULL    |       |
| product_name | varchar(30) | YES  |     | NULL    |       |
| price       | int(11)     | YES  |     | NULL    |       |
| customer_id | int(11)     | YES  | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
```

Now let update customer table,

```
MariaDB [customer_db]> update customers set name='gita' where customer_id=3;
Query OK, 1 row affected (0.007 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [customer_db]> select * from customers;
+-------------+-------+----------------+------+-----------+
| customer_id | name  | email          | age  | address   |
+-------------+-------+----------------+------+-----------+
|           1 | ram   | ram@gmail.com  |   22 | kathmandu |
|           2 | sita  | sita@gmail.com |   25 | pokhara   |
|           3 | gita  | gita@gmail.com |   26 | butwal    |
|           4 | hari  | hari@gmail.com |   27 | lalitpur  |
|           5 | gopal | gopal@gmail.com |   20 | bhaktapur |
+-------------+-------+----------------+------+-----------+
```

Again insert some rows in orders table

```
MariaDB [customer_db]> insert into orders values(1,'iron',1200,1);
Query OK, 1 row affected (0.007 sec)

MariaDB [customer_db]> insert into orders values(2,'guitar',4500,2);
Query OK, 1 row affected (0.008 sec)

MariaDB [customer_db]> insert into orders values (3,'phone',57000,1);
Query OK, 1 row affected (0.008 sec)

MariaDB [customer_db]> insert into orders values(4,'printer',88000,3);
Query OK, 1 row affected (0.007 sec)

MariaDB [customer_db]> insert into orders values(5,'drone',125000,2);
Query OK, 1 row affected (0.008 sec)

MariaDB [customer_db]> insert into orders values(6,'smartwatch',7500,4);
Query OK, 1 row affected (0.007 sec)

MariaDB [customer_db]> insert into orders values(7,'oven',14500,5);
Query OK, 1 row affected (0.007 sec)
```

```
MariaDB [customer_db]> select * from orders;
+----------+--------------+--------+-------------+
| order_id | product_name | price  | customer_id |
+----------+--------------+--------+-------------+
|        1 | iron         |   1200 |           1 |
|        2 | guitar       |   4500 |           2 |
|        3 | phone        |  57000 |           1 |
|        4 | printer      |  88000 |           3 |
|        5 | drone        | 125000 |           2 |
|        6 | smartwatch   |   7500 |           4 |
|        7 | oven         |  14500 |           5 |
+----------+--------------+--------+-------------+
```

```
MariaDB [customer_db]> select * from customers;
+-------------+-------+----------------+------+----------+
| customer_id | name  | email          | age  | address  |
+-------------+-------+----------------+------+----------+
|           1 | ram   | ram@gmail.com  |   22 | kathmandu |
|           2 | sita  | sita@gmail.com |   25 | pokhara  |
|           3 | gita  | gita@gmail.com |   26 | butwal   |
|           4 | hari  | hari@gmail.com |   27 | lalitpur |
|           5 | gopal | gopal@gmail.com |  20 | bhaktapur |
+-------------+-------+----------------+------+----------+
```

Try to insert customer_id in orders table that does not exist in customers table

```
MariaDB [customer_db]> insert into orders values(8,'Laptop',158000,6);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`customer_db`.`orders`, CONSTRAINT `custorder_fk` FOREIGNKEY (`customer_id`) REFERENCES
`customers` (`customer_id`))
MariaDB [customer_db]>
```

This cannot be inserted, because it violates foreign key consraints.